Cool math is hiding in "The 12 Days of Christmas"

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1 Introduction

The classic English Christmas carol "The Twelve Days of Christmas" has all kinds of interesting mathematical features. For example, it is an example of a *cumulative song*, in which the lyrics detail a series of increasingly numerous gifts given to the speaker by their "true love" on each of the twelve days of Christmas (the twelve days that make up the Christmas season, starting with Christmas Day) [6]. This cumulative property leads to a bunch of interesting questions including "How many gifts are given each day?" and "What is the total number of gifts given during the 12 Days of Christmas?".

In any event the carol's words were apparently first published in England in the late 18th century (and has a Roud Folk Song Index number of 68 if you happen to be interested in that kind of thing [8]). A large number of different melodies have been associated with the song, of which the best known is derived from a 1909 arrangement of a traditional folk melody by English composer Frederic Austin [9]. There are twelve verses, each describing a gift given by "my true love" on one of the twelve days of Christmas. Suffice it to say that are many variations in the lyrics. That said, the lyrics I'm using here are from Frederic Austin's 1909 publication that established the current form of the carol [7].

The first three verses are:

1. On the first day of Christmas my true love sent to me: *A partridge in a pear tree.*
2. On the second day of Christmas my true love sent to me: *Two turtle doves, and a partridge in a pear tree.*
3. On the third day of Christmas my true love sent to me: *Three French hens, two turtle doves, and a partridge in a pear tree.*

The subsequent verses follow the same pattern. The verses deal with the next day where it adds one new gift and then repeats all the earlier gifts. So each verse is one line longer than the previous verse. The subsequent verses add

4. 4 calling birds
5. 5 gold rings
6. 6 geese a-laying
7. 7 swans a-swimming
8. 8 maids a-milking
9. 9 ladies dancing
10. 10 lords a-leaping
11. 11 pipers piping
12. 12 drummers drumming

2 How many gifts are received each day and what is the total?

The number of gifts received on day **n** and the total number of gifts received by (including) day **n** are shown in the following table ($1 \leq \mathbf{n} \leq 12$):

Day	Gifts received on day n	Total number of gifts received by day n
1	**1**	**1**
2	1+2=**3**	1+3=**4**
3	1+2+3=**6**	1+3+6=**10**
4	1+2+3+4=**10**	1+3+6+10=**20**
5	1+2+3+4+5=**15**	1+3+6+10+15=**35**
6	1+2+3+4+5+6=**21**	1+3+6+10+15+21=**56**
7	1+2+3+4+5+6+7=**28**	1+3+6+10+15+21+28=**84**
8	1+2+3+4+5+6+7+8=**36**	1+3+6+10+15+21+28+36=**120**
9	1+2+3+4+5+6+7+8+9=**45**	1+3+6+10+15+21+28+36+45=**165**
10	1+2+3+4+5+6+7+8+9+10=**55**	1+3+6+10+15+21+28+36+45+55=**220**
11	1+2+3+4+5+6+7+8+9+10+11=**66**	1+3+6+10+15+21+28+36+45+55+66=**286**
12	1+2+3+4+5+6+7+8+9+10+11+12=**78**	1+3+6+10+15+21+28+36+45+55+66+78=**364**

Table 1: Gift count for the 12 days of Christmas

Interestingly, the numbers in blue in Table 1, the number of gifts received on day **n**, are called triangular numbers. Why? We will see in Section 3 that there is an interesting geometric interpretation that motivates this name.

The nth triangular number, T_n, is defined as follows:

$$T_n = \sum_{k=1}^{n} k = \frac{n(n+1)}{2} = \binom{n+1}{2} \tag{1}$$

where $\binom{n}{m}$ is the familiar binomial coefficient[1]. For example, $T_{12} = \sum_{k=1}^{12} k = \frac{12 \cdot 13}{2} = \binom{13}{2} = \mathbf{78}$.

So how many total gifts have been received by day **n**? If we look at the **Total number of gifts received by day n** column of Table 1 we can see a pattern. In particular, the number of gifts received by (including) day **n** is the sum of the gifts received on the previous days plus the gift received on day **n**. The number of gifts received by day **n** is shown in red in Table 1. These numbers are called the tetrahedral numbers and are defined as follows:

$$T_{e_n} = \sum_{k=1}^{n} T_k = \frac{n(n+1)(n+2)}{6} = \binom{n+2}{3} \tag{2}$$

where T_k is the kth triangular number (Equation (1)).

It turns out that the total number of gifts received by day 12, $T_{e_{12}}$, is $\sum_{k=1}^{12} T_k = \frac{12 \cdot 13 \cdot 14}{6} = \binom{14}{3} = \mathbf{364}$.

[1] Aside: The great mathematician Carl Friedrich Gauss is said to have discovered the consecutive integer formula (Equation (1)) while still at primary school [2].

BTW, can a number can be both triangular and tetrahedral? Well, if so there would have to exist $m, n \in \mathbb{Z}^+$ such that

$$T_n = \binom{n+1}{2} = \binom{m+2}{3} = T_{e_m} \tag{3}$$

Apparently the only m, n that satisfy Equation (3) are [3]:

$$\begin{aligned}
T_{e_1} &= T_1 &&= 1 \\
T_{e_3} &= T_4 &&= 10 \\
T_{e_8} &= T_{15} &&= 120 \\
T_{e_{20}} &= T_{55} &&= 1540 \\
T_{e_{34}} &= T_{119} &&= 7140
\end{aligned}$$

Also interesting is the connection to Pascal's triangle [4]. Specifically, the 3rd diagonal row of Pascal's triangle, shown in blue in Figure 1, contains the numbers we see in **Gifts received on day n** column of Table 1. These are the triangular numbers (Equation (1)). Similarly, the 4th diagonal row of Pascal's triangle, shown in red in Figure 1, contains the **Total number of gifts received by day n** column of Table 1. These are the tetrahedral numbers (Equation (2)).

1														
1	1													
1	2	1												
1	3	3	1											
1	4	6	4	1										
1	5	10	10	5	1									
1	6	15	20	15	6	1								
1	7	21	35	35	21	7	1							
1	8	28	56	70	56	28	8	1						
1	9	36	84	126	126	84	36	9	1					
1	10	45	120	210	252	210	120	45	10	1				
1	11	55	165	330	462	462	330	165	55	11	1			
1	12	66	220	495	792	924	792	495	220	66	12	1		
1	13	78	286	715	1287	1716	1716	1287	715	286	78	13	1	
1	14	91	364	1001	2002	3003	3432	3003	2002	1001	364	91	14	1

Figure 1: Pascal's Triangle with the triangular numbers in **blue** and the tetrahedral numbers in **red**

3 Geometric interpretations

There are interesting geometric interpretations of both the triangular numbers and the tetrahedral numbers. So first, what exactly is a triangular number? The first six triangular numbers are depicted as triangles in Figure 2. We can see from the figure that a triangular number T_n is the sum of the integers from 1 to n [5]. This is also the number of gifts received on the nth day of Christmas according to the "12 Days of Christmas".

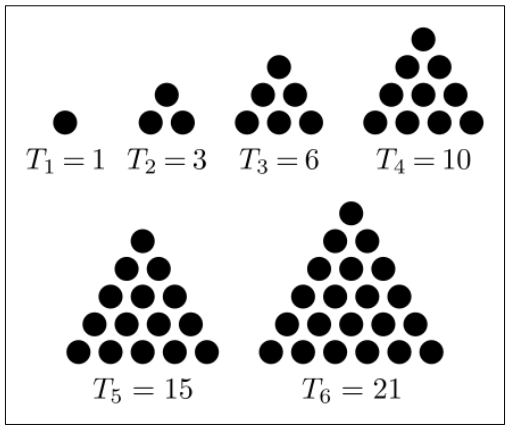

Figure 2: The First Six Triangular Numbers

Next, what are the tetrahedral numbers? As we saw in Equation (2), the nth tetrahedral number is the sum of the first n triangular numbers [1]. If we stack up these triangular numbers we get a tetrahedron (hence the name). For example, T_{e_4} is depicted in Figure 3. The nth tetrahedral number turns out to be the number of gifts received up to and including day **n** in "The 12 Days of Christmas".

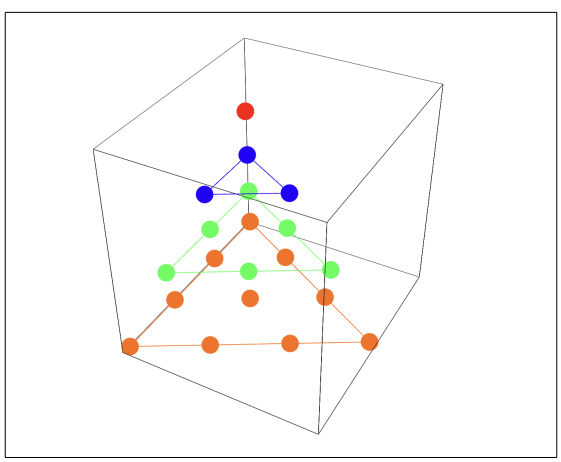

Figure 3: The Tetrahedral Numbers

LaTeX Source

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